2024 8th International Symposium on Computer Science and Intelligent Control (ISCSIC) | 979-8-3503-8028-6/24/$31.00 ©2024 IEEE | DOI: 10.1109/ISCSIC64297.2024.00011

Fault-tolerant Control of Variable Load Drone Based on HOFA System Approach

1st Runze Wang
Department of Control Sciences and Engineering
Harbin Institute of Technology
Harbin, China
23B904007@stu.hit.edu.cn

2nd Ruizhi Tong
Department of Control Sciences and Engineering
Harbin Institute of Technology
Harbin, China
23S104097@stu.hit.edu.cn

3rd Linjie Li
Department of Control Sciences and Engineering
Harbin Institute of Technology
Harbin, China
23S136183@stu.hit.edu.cn

4th Yi Zeng*
Department of Control Sciences and Engineering
Harbin Institute of Technology
Harbin, China
yi.zeng@hit.edu.cn

***Abstract*—This paper proposes an integral sliding controller based on second-order fully actuated system theory for the control problem of variable load six rotor unmanned aerial vehicles in the event of flight faults, and an extended observer with model information is used to estimate and compensate for the fault signals which occur during the flight. This paper innovatively introduces virtual control variables to supplement the system into a second-order six dimensional fully actuated system, and designs an integral sliding mode controller using a modified saturation function to achieve rapid stability of unmanned aerial vehicles during the process of verying load. In addition, the extended observer can also successfully estimate and compensate for the loss of execution efficiency of the drone.**

***Index Terms*—High-order Fully Actuated Systems, Various Load Drones, Sliding Mode control, Fault-Tolerant Control**

I. Introduction

A great number of unmanned aerial vehicles (UAVs) have been successfully applied in many different fields. For example transportation and agriculture over the last decade. They can not only free people from repetive labor, but also complete some dangerous works such as disaster recovery and moutain mapping. Different works under various situation have put forward diverse requirements for drones, including the ability to carry various loads and cope with rotor fault events. The controllers, as the core of UAVs, need to adopt advanced methods with strong robustness and disturbance resistance to deal with emerging problems. Numerous attitude control methods are proposed, considering the following control approach: proportional-integral-derivative (PID) control [1], linear-quadratic regulator (LQR) [2], model predictive control (MPC) [3], and sliding mode control (SMC) [4], High-Order Fully Actuated (HOFA) system theory [5], etc.

This work was supported in part by Harbin Institute of Technology, and in part by the National Natural Science Foundation of China under Grant 62188101, 62303135,62033005, and 62320106001, and in part by the Natural Science Foundation of Heilongjiang Province (ZD2021F001), and in part by the Heilongjiang Touyan Team Program. (corresponding author: Yi Zeng)

The algorithms mentioned above have their own features. H.Bolandi [1] combined a PID controller with a genetic algorithm to implement the stability of quadcopters. In order to make the system respond faster, M.R.Cohen [2] researched optimal control for a quadcopter drone based on discrete-time finite-horizon LQR, demonstrating superior performance compared to traditional LQR controllers. Although MPC has difficulties in tracking complex trajectories, M.Islam [3] designed a MPC controller exploiting a linear modelfor prediction. The results indicated its good dynamic performance and tracking ability. To solve uncertainty and external disturbance problems and improve the robustness of the system, Abro, G.E.M [4] introduced a model-free one-dimensional fuzzy sliding mode control (MFSDF-SMC) to control the attitude and position of underactuated UAV.

In recent years, the Fully Actuated System (FAS) approach [5] has attracted attention of the academia. This approach exhibits excellent characteristics, as fully actuated systems are commonly present in the real world, and many systems' initial mathematical models can be described as natural second-order or higher-order fully actuated systems. This form of systems is relatively simple and possesses full actuation characteristics, which reduces the complexity of controller design [5]. Marko Pranjic [6] successfully proved that this theory can be applied to multi rotor systems and compared and analyzed the advantages and disadvantages of traditional control and LQR. Lu [7] proposed an improved HOFA method which divided the system into two subsystems and displayed its effectiveness in his work. Yan [8] combined this theory with disturbance observer (DOB) and enabled a quadcopter affected by external interference to stably track a designated trajectory. DOB was also used by Li [9] to form a novel fault-tolerant control with HOFA and the results illustrated that this controller was able to deal with actuator faults.

Although dozens of papers have shown their researches on HOFA and UAV system, there still are tons of cases which deserved to be studied. Lots of works, it should be noted, still

979-8-3503-8028-6/24/$31.00 ©2024 IEEE
DOI 10.1109/ISCSIC64297.2024.00011

Authorized licensed use limited to: Netaji Subhas University of Technology New Delhi. Downloaded on June 13,2025 at 04:37:07 UTC from IEEE Xplore. Restrictions apply.

focus on traditional control methods ignoring model information, because coupling and nonlinearity are considered as the model uncertainty. Meanwhile, different application scenarios have various functional requirements for drones, for example extra load. This work focuses on in-depth research on fault-tolerant algorithms for variable load unmanned aerial vehicles. We propose an improved integral sliding mode controller based on HOFA system theory to eliminate the impact of load changes and add an model-based extended observer into the controller to handle fault events. The results show that this method has strong robustness, and can effectively handle the load changes and fault events of drones during flight.

II. High-order Fully Actuated System Model for Variable Load Drone

According to the description in [10], we can obtain a general model of rotor unmanned aerial vehicles (UAV) :

$$\begin{cases} \ddot{x} = -\frac{K_x}{m_b}\dot{x} + \frac{\cos\phi\sin\theta\cos\psi+\sin\phi\sin\psi}{m_b}F_z \\ \ddot{y} = -\frac{K_y}{m_b}\dot{y} + \frac{\cos\phi\sin\theta\sin\psi-\sin\phi\cos\psi}{m_b}F_z \\ \ddot{z} = -\frac{K_z}{m_b}\dot{z} - g + \frac{\cos\phi\cos\theta}{m_b}F_z \\ \ddot{\phi} = \frac{I_{by}-I_{bz}}{I_{bx}}\dot{\theta}\dot{\psi} + \frac{1}{I_{bx}}\tau_\phi \\ \ddot{\theta} = \frac{I_{bz}-I_{bx}}{I_{by}}\dot{\phi}\dot{\psi} + \frac{1}{I_{by}}\tau_\theta \\ \ddot{\psi} = \frac{I_{bx}-I_{by}}{I_{bz}}\dot{\phi}\dot{\theta} + \frac{1}{I_{bz}}\tau_\psi \end{cases}, \tag{1}$$

where x, y and z are three components of drone position in the ground coordinate system; ϕ, ψ and θ are the attitude angle of drones. Therefore, $\dot{i}$ and $\ddot{i}$ $(i = x, y, z, \phi, \psi, \theta)$ mean the first and second derivatives of i. Meanwhile, I_{bx}, I_{by} and I_{bz} represent the rotational inertia of the drone in three axial directions; K_x, K_y and K_z are the air resistance coefficients in three directions; m_b is the mass of UAV and F_z is the lift force provided by rotors.

When adding additional loads to a drone, multiple factors need to be considered, such as whether the connection between drones and loads is rigid, and the impact of changes in centroidal position on the system. In order to simplify the model in this paper, we assume that the mass distribution of the load is uniform and symmetrical; loads are rigidly connected to UAV; after the drone carries goods, centroidal position of goods is horizontally projected at the same position as the drone's center of mass. Let m_s represent the mass of the load; $r_s = [0, 0, z_s]^{\mathrm{T}}$ represent the distance from the barycenter location of the load to the coordinate origin in aircraft-body coordinate frame, where z_s represent the component of r_s along the z-axis in the body coordinate system; $I_s = [I_{sx}, I_{sy}, I_{sz}]^{\mathrm{T}}$ represents the moment of inertia for goods, where I_{sx}, I_{sy}, I_{sz} represents the components of I_s on the three-axis direction, and the mass of loaded drone system m can be obtained:

$$m = m_b + m_s. \tag{2}$$

The inertia of drone carrying cargo in three directions:

$$\begin{cases} I_x = I_{bx} + I_{sx} + m_s z_s^2 \\ I_y = I_{by} + I_{sy} + m_s z_s^2 \\ I_z = I_{bz} + I_{sz} \end{cases}. \tag{3}$$

Ignoring the impact caused by changes in the centroid position, a simplified model of the loaded UAV system can be obtained as (4):

$$\begin{cases} \ddot{x} = -\frac{K_x}{m}\dot{x} + \frac{\cos\phi\sin\theta\cos\psi+\sin\phi\sin\psi}{m}F_z \\ \ddot{y} = -\frac{K_y}{m}\dot{y} + \frac{\cos\phi\sin\theta\sin\psi-\sin\phi\cos\psi}{m}F_z \\ \ddot{z} = -\frac{K_z}{m}\dot{z} - g + \frac{\cos\phi\cos\theta}{m}F_z \\ \ddot{\phi} = \frac{I_y-I_z}{I_x}\dot{\theta}\dot{\psi} + \frac{\tau_\phi}{I_x} \\ \ddot{\theta} = \frac{I_z-I_x}{I_y}\dot{\phi}\dot{\psi} + \frac{\tau_\theta}{I_y} \\ \ddot{\psi} = \frac{I_x-I_y}{I_z}\dot{\phi}\dot{\theta} + \frac{\tau_\psi}{I_z} \\ m = m_b + m_s \\ I_x = I_{bx} + I_{sx} + m_s z_s^2 \\ I_y = I_{by} + I_{sy} + m_s z_s^2 \\ I_z = I_{bz} + I_{sz} \end{cases}. \tag{4}$$

According to Duan [11] [12], a second-order fully actuated system (8) can be obtained:

$$E\ddot{x} = f(x, \dot{x}, t) + B(x, \dot{x}, t)u - v, \tag{5}$$

where the control distribution matrix satisfies the following fully actuated conditions:

$$\det B(x, x, t) \neq 0, \forall x, \dot{x} \in R^n, t \geq 0. \tag{6}$$

When the system satisfies the fully actuated conditions, for any given matrix $A_1, A_0 \in R^{n\times n}$, it can always be achieved through the control law:

$$u = -B^{-1}(\cdot)[A_1\dot{x} + A_0 x + f(x, \dot{x}, t) - v]. \tag{7}$$

Obtain the following linear steady-state closed-loop system:

$$E\ddot{x} + A_1\dot{x} + A_0 x = \nu. \tag{8}$$

We define

$$\begin{cases} u_1 = \frac{\cos\phi_d\sin\theta_d\cos\psi_d+\sin\phi_d\sin\psi_d}{m+\Delta m}F_z \\ u_2 = \frac{\cos\phi_d\sin\theta_d\sin\psi_d-\sin\phi_d\cos\psi_d}{m+\Delta m}F_z \\ u_3 = \frac{\cos\phi_d\cos\theta_d}{m+\Delta m}F_z \\ u_4 = \frac{1}{I_x}\tau_\phi \\ u_5 = \frac{1}{I_y}\tau_\theta \\ u_6 = \frac{1}{I_z}\tau_\psi \end{cases}. \tag{9}$$

Among them, φ_d, θ_d, ϕ_d represents the desired angle. Subsequently, we can obtain:

$$E\ddot{x} = f(x, \dot{x}, t) + B(x, \dot{x}, t)u - v, \tag{10}$$

where E and B are both the identity matrices of $I_{6\times 6}$

$$f(x, \dot{x}, t) = \mathrm{diag}(-\frac{K_1}{m+\Delta m}\dot{x}, -\frac{K_2}{m+\Delta m}\dot{y}, -\frac{K_3}{m+\Delta m}\dot{z}, \frac{I_y-I_z}{I_x}\dot{\theta}\dot{\psi}, \frac{I_z-I_x}{I_y}\dot{\phi}\dot{\psi}, \frac{I_x-I_y}{I_z}\dot{\phi}\dot{\theta}), \tag{11}$$

$$v = \begin{bmatrix} 0 & 0 & g & 0 & 0 & 0 \end{bmatrix}^{\mathrm{T}}. \tag{12}$$

Authorized licensed use limited to: Netaji Subhas University of Technology New Delhi. Downloaded on June 13,2025 at 04:37:07 UTC from IEEE Xplore. Restrictions apply.

III. Sliding Mode Control

The sliding mode control proposed in the last century is a simple, feasible and effective control algorithm with strong robustness. In this chapter, an improved integral sliding mode controller based on the fully actuated system theory will be proposed.

The error of state variables e_1 and its first derivative are defined as [13]:

$$e_1 = x - x_{\mathrm{d}}. \tag{13}$$

$$e_2 = \dot{x} - \dot{x}_d, \tag{14}$$

where x represents the state variable and x_{d} is the expected input for the state variable. Thus the general formula of the integral sliding mode control can be given:

$$s = e_2 + k_{\mathrm{p}} e_1 + k_{\mathrm{i}} \int_0^t e_1 \mathrm{d}\tau, \tag{15}$$

where k_{p} and $k_i$are two positive constants.

A nonlinear fuction H is introduced to replace the original integral term, which can deal with integral saturation problem. Meanwhile in order to reduce the overshoot, another positive parameter k_d is introduced. The new formula is as follows:

$$\begin{cases} s = k_d e_2 + k_p e_1 + k_i H - e_2(0) - k_p e_1(0) \\ \dot{H} = h(e) \end{cases}, \tag{16}$$

where:

$$H(e) = \begin{cases} \frac{2\beta}{\pi}(1 - \cos(\frac{\pi}{2\beta}e)) & |e| < \beta \\ \beta_e - \frac{\pi - 2}{\pi}\beta^2 & e \geq \beta \\ -\beta_e - \frac{\pi - 2}{\pi}\beta^2 & e \leq -\beta \end{cases}. \tag{17}$$

In the fomula above, β is a positive constant meaning switch boundaries. Taking the derivative of $H(e)$, we obtain a new nonlinear function h:

$$h(e) = \begin{cases} \beta \sin \frac{\pi \rho}{2\beta} & |e| < \beta \\ \beta & e \geq \beta \\ -\beta & e \leq -\beta \end{cases}. \tag{18}$$

Therefore, the derivative of fomula (16) above is obtained:

$$\begin{cases} \dot{s} = k_d \dot{e}_2 + k_p e_2 + k_i h(e) \\ \dot{H} = h(e) \end{cases}. \tag{19}$$

Combining (19) and HOFA model (10), we can obtain:

$$\begin{aligned} &\dot{s} = k_{\mathrm{d}} \left(\ f(x, \dot{x}, t) + B(x, \dot{x}, t)u - v - \ddot{x}_d \ \right) \\ &+ k_{\mathrm{p}} e_2 + k_{\mathrm{i}} h(e) \end{aligned}. \tag{20}$$

For the approach law, we adopt the traditional exponential approach law here:

$$\dot{s} = -\varepsilon \mathrm{sigm(s)} - ks \quad k \geq 0, \varepsilon \geq 0. \tag{21}$$

In the law, $\varepsilon \mathrm{sigm(s)}$ is the constant speed reaching term, k_s is the exponential reaching term. Usually, k is designed to be a large value, and ε is designed to be a small value. When the system state is far from the sliding surface, the exponential reaching term can make the system state quickly approach the vicinity of the sliding surface. When the system state is close to the sliding surface, the constant speed reaching term plays a major role. At this time, the approaching speed is ε, ensuring that the system state reaches the sliding surface at a relatively small speed.

We define $\mathrm{sigm(s)} = \left(\frac{1}{1 - e^{-bs}} - 0.5 \right)$, in which b represents the convergence speed.

Combining (20) and (21), we can obtain:

$$\begin{aligned} &k_{\mathrm{d}} \left(\ f + Bu - v - \ddot{x}_d \ \right) + k_{\mathrm{p}} e_2 + k_{\mathrm{i}} h(e) \\ &+ \varepsilon \mathrm{sigm(s)} + ks = 0 \end{aligned}. \tag{22}$$

Therefore, the control law for the integral sliding mode is obtained:

$$u = \frac{1}{B}(-f + v + \ddot{x}_d - \frac{k_{\mathrm{p}} e_2}{k_{\mathrm{d}}} - \frac{k_{\mathrm{i}} h(e)}{k_{\mathrm{d}}} - \frac{\varepsilon \mathrm{sigm(s)}}{k_{\mathrm{d}}} - \frac{ks}{k_{\mathrm{d}}}). \tag{23}$$

Next we proof the stability using Lyapunov function which is given as follow:

$$V = \frac{1}{2} s^2. \tag{24}$$

$$\begin{aligned} &\dot{V} = s\dot{s} = s \cdot (k_d \dot{e}_2 + k_P e_2 + k_i h(e)) = \\ &s(k_{\mathrm{d}}(f + Bu - v - \ddot{x}_d) + k_{\mathrm{p}} e_2 + k_{\mathrm{i}} h(e)) \end{aligned}. \tag{25}$$

Substituting the expression for the control input u in (23), we get:

$$\dot{V} = k_{\mathrm{d}} s(-\varepsilon \mathrm{sigm(s)} - ks) = -k_1 s \varepsilon \mathrm{sigm(s)} - k_2 s^2. \tag{26}$$

In general, we set k_1 and k_2 to positive values, which means that $\dot{V}$ will be negative definite when the value of s is non-zero. Therefore, it can be determined that the system is stabilized globally.

IV. Fault-tolerant observer design

Due to actuator malfunctions, such as propeller damage, the execution efficiency may deviate either towards high or low values. This aspect can be considered as disturbances in the system, and compensation can be achieved by constructing an observer for the system. A core component of the self-disturbance rejection control is the Extended State Observer (ESO), which effectively suppresses the influence of external uncertainties on the system. The fundamental idea of this approach is to introduce disturbances that affect the controlled output into new state variables and suppress these disturbances. This process linearizes the system, transforming it into a cascaded system, ultimately controlled using an integrator.

$$\begin{cases} \ddot{x} = -\frac{K_1}{m + \Delta m}\dot{x} + \frac{\cos\phi \sin\theta \cos\psi + \sin\phi \sin\psi}{m + \Delta m} F_z + d_1 \\ \ddot{y} = -\frac{K_2}{m + \Delta m}\dot{y} + \frac{\cos\phi \sin\theta \sin\psi - \sin\phi \cos\psi}{m + \Delta m} F_z + d_2 \\ \ddot{z} = -\frac{K_3}{m + \Delta m}\dot{z} - g + \frac{\cos\phi \cos\theta}{m + \Delta m} F_z + d_3 \\ \ddot{\phi} = \frac{I_y - I_z}{I_x}\dot{\theta}\dot{\psi} + \frac{1}{I_x}\tau_\phi + d_4 \\ \ddot{\theta} = \frac{I_z - I_x}{I_y}\dot{\phi}\dot{\psi} + \frac{1}{I_y}\tau_\theta + d_5 \\ \ddot{\psi} = \frac{I_x - I_y}{I_z}\dot{\phi}\dot{\theta} + \frac{1}{I_z}\tau_\psi + d_6 \end{cases}. \tag{27}$$

Authorized licensed use limited to: Netaji Subhas University of Technology New Delhi. Downloaded on June 13,2025 at 04:37:07 UTC from IEEE Xplore. Restrictions apply.

In the context of n-order nonlinear time-varying systems,

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = x_3 \\ \vdots \\ \dot{x}_n = f\left(x_1, x_2, \cdots, x_n, w(t), t\right) + bu \\ y = x_1 \end{cases} . \tag{28}$$

The expanded state of the system can be represented as follows:

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = x_3 \\ \vdots \\ \dot{x}_n = x_{n+1} + bu \\ \dot{x}_{n+1} = f(t) \\ y = x_1 \end{cases} . \tag{29}$$

The estimated values of the expanded system state variables are designated as: $Z = \begin{bmatrix} z_1 & z_2 & \cdots & z_n & z_{n+1} \end{bmatrix}^{\mathrm{T}}$, state observer can be established as follows:

$$\begin{cases} e = z_1 - y \\ \dot{z}_1 = z_2 - \boldsymbol{\beta}_1 e \\ \dot{z}_2 = z_3 - \boldsymbol{\beta}_2 \mathrm{fal}\big(e, \alpha_1, \delta\big) \\ \vdots \\ \dot{z}_n = z_{n+1} - \boldsymbol{\beta}_n \mathrm{fal}\big(e, \alpha_{n-1}, \delta\big) + bu \\ \dot{z}_{n+1} = -\boldsymbol{\beta}_{n+1} \mathrm{fal}\big(e, \alpha_n, \delta\big) \end{cases} . \tag{30}$$

The control gain is represented by b, the state-space expressions for each control channel are as follows:

$$\begin{cases} \dot{x_1} = x_2 \\ \dot{x_2} = f(x_2 + 1) + bu + d \\ y = x_1 \end{cases} . \tag{31}$$

The expanded state of the system can be represented as follows:

$$\begin{cases} x_1 = x_2 \\ x_2 = f(x_2, t) + bu + x_3 \\ x_3 = d(t) \\ y = x_1 \end{cases} . \tag{32}$$

The estimated values of the expanded system state variables are designated as: $Z = \begin{bmatrix} z_1 & z_2 & z_3 \end{bmatrix}^{\mathrm{T}}$, state observer can be established as follows:

$$\begin{cases} e = z_1 - y \\ z_1 = z_2 - \beta_1 g_1(e) \\ z_2 = z_3 - \beta_2 g_2(e) + 6u + f(x_2, t) \\ z_3 = -\beta_3 g_3(e) \end{cases} . \tag{33}$$

TABLE I
THE MODEL PARAMETER VALUES

Parameters	value
m	17kg
Δm	1kg
k_1	0.001N/$(s \cdot m^2)$
k_2	0.001N/$(s \cdot m^2)$
k_3	0.003N/$(s \cdot m^2)$
I_x	0.2kg·m^2
I_y	0.2kg·m^2
I_z	0.37kg·m^2

TABLE II
THE GAIN SELECTION OF THE ATTITUDE CONTROLLER

Parameters	x	y	z	φ	θ	ϕ
k_d	2	2	2	0.5	0.5	0.5
k_p	50	50	50	80	80	80
k_i	80	80	80	100	100	40
k	10	10	10	10	10	10
ε	0.1	0.1	0.1	1	1	1
β	10	10	10	10	10	10
a	10	10	10	10	10	10

V. Numerical Example

A numerical example is conducted on the sine wave and ramp signal response of the system to verify its effectiveness. Table I and II show the values of the model parameters as well as the controller respectively.

Fig. 1 and 2 respectively show the position information and tracking error of the drone. From Fig. 2, we can see that it can quickly converge after tracking y and z for 0.3 seconds. When tracking x, the error approaches zero within 0.3 seconds and fluctuates around the zero scale line, with an up and down error of less than 0.005 meters. The position error is controlled within a small range, indicating that the designed controller has good tracking performance and can reach the specified position in a short time.

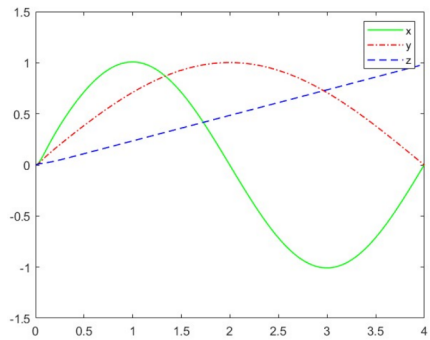

Fig. 1. The response of the position in x, y and z

Fig. 3 and 4 respectively show the expected attitude and tracking attitude error of the drone. From Fig. 4., we can see that the tracking attitude angle in ϕ, φ and θ converge quickly within 0.5 seconds and there is no overshoot or oscillation.

Authorized licensed use limited to: Netaji Subhas University of Technology New Delhi. Downloaded on June 13,2025 at 04:37:07 UTC from IEEE Xplore. Restrictions apply.

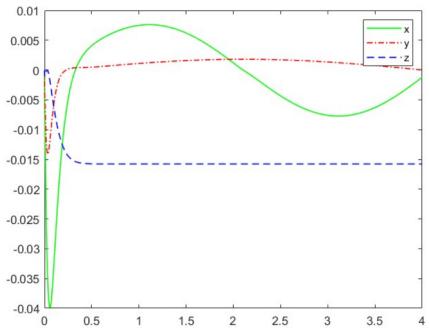

Fig. 2. The tracking error of the position in x, y and z

Moreover, the impact of adding a load on the tracking error of the attitude angle is minimal, with a maximum error of no more than 0.2 degrees. The attitude angle error is controlled within a small range, indicating that the controller has high control accuracy and good performance.

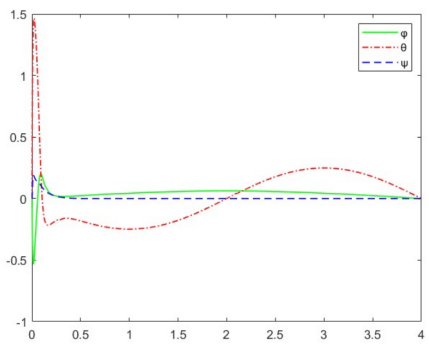

Fig. 3. The response of the angle in roll φ, pitch θ and yaw axes ϕ

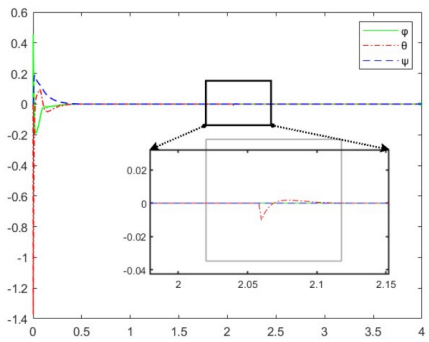

Fig. 4. The tracking error of the angle in roll φ, pitch θ and yaw axes ϕ

VI. Conclusion

This article proposes an integral sliding mode controller based on second-order fully actuated system theory for the control problem of variable load unmanned aerial vehicles in the event of flight faults. An extended observer with model information is used to estimate and compensate for the fault signals. Finally, a numerical example is used to verify the effectiveness of the method and the results indicate that the controller offers the strong robustness and excellent dynamic performance. In the future, more consideration will be given to the actual mission requirements of the aircraft, and further research on aircraft control based on the HOFA system method will be carried out.

References

[1] Bolandi H, Rezaei M, Mohsenipour R, et al. Attitude control of a quadrotor with optimized PID controller[J]. 2013.

[2] Cohen M R, Abdulrahim K, Forbes J R. Finite-horizon LQR control of quadrotors on $SE_2(3)$[J]. IEEE Robotics and Automation Letters, 2020, 5(4): 5748-5755.

[3] Islam M, Okasha M, Idres M M. Dynamics and control of quadcopter using linear model predictive control approach[C]//IOP conference series: materials science and engineering. IOP Publishing, 2017, 270(1): 012007.

[4] Abro G E M, Zulkifli S A B M, Asirvadam V S, et al. Model-free-based single-dimension fuzzy SMC design for underactuated quadrotor UAV[C]//Actuators. MDPI, 2021, 10(8): 191.

[5] G.R.Duan. Pseudo-Linear System Approach to Aircraft Control - Part II: Methods and Prospects [J]. Journal of Astronautics,2020,41(7): 839-849.

[6] Pranjic M, Pender A, Piljek P, et al. Fully Actuated Multirotor UAV Control Design and Implementation[C]//2022 International Conference on Electrical, Computer and Energy Technologies (ICECET). IEEE, 2022: 1-6.

[7] Lu S, Tsakalis K, Chen Y. Development and application of a novel high-order fully actuated system approach-Part I: 3-DOF quadrotor control[J]. IEEE Control Systems Letters, 2022, 7: 1177-1182.

[8] Yan X, Yu L, Shao G, et al. Robust Trajectory Tracking Control for Quadrotor UAV with External Disturbances via Fully Actuated System Approach[C]//2023 2nd Conference on Fully Actuated System Theory and Applications (CFASTA). IEEE, 2023: 783-787.

[9] Li M, Zhang K, Miao Q, et al. Adaptive Fault-Tolerant Control for Attitude and Altitude Synchronization of Quadrotors Based on Fully-Actuated System Approaches[C]//2023 2nd Conference on Fully Actuated System Theory and Applications (CFASTA). IEEE, 2023: 480-485.

[10] Rajappa S, Ryll M, Bulthoff H H, et al. Modeling, control and design optimization for a fully-actuated hexarotor aerial vehicle with tilted propellers[C]//2015 IEEE international conference on robotics and automation (ICRA). IEEE, 2015: 4006-4013.

[11] G.R.Duan. High-order system methods I: Fully actuated systems and parametric design [J]. Acta Automatica Sinica,2020,46(7): 1333-1345.

[12] Zhao Q, Duan G R. Fully actuated system approach for 6DOF spacecraft control based on extended state observer[J]. Journal of Systems Science and Complexity, 2022, 35(2): 604-622.

[13] Chen G X, He Y B, Qiu G F, et al. Integral Sliding Mode Control of Permanent Magnet Synchronous Linear Motor [J]. Modular Machine Tool and Automatic Manufacturing Technique, 2023(10): 77-80.

Authorized licensed use limited to: Netaji Subhas University of Technology New Delhi. Downloaded on June 13,2025 at 04:37:07 UTC from IEEE Xplore. Restrictions apply.